

Solving with the Quadratic Formula

Your Turn — solve a quadratic, one step at a time

Name _____

Date _____

Period _____

1 Set the equation equal to 0



2 Identify a , b , and c



3 Write the quadratic formula

$$X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



4 Substitute a , b , c — keep the parentheses!



5 Simplify under the radical



6 Take the square root (look for a perfect square)



7 Split into two equations



8 Simplify each

✓ Check Your Answer

Factor, or substitute each answer back into the original equation.

★ Remember

- Set it = 0 before you start.
- Keep parentheses when you substitute — watch negatives.
- Look for a perfect square under the radical.
- Not a perfect square? The answer is irrational — leave it in exact (radical) form.

SOLUTIONS

$X =$ _____ or $X =$ _____